



Osman Talayhan

Software Developer

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ABOUT ME

I am a backend-oriented software developer with a strong focus on algorithms, distributed systems, and engineering applied to artificial intelligence, particularly in computer vision. I primarily work on designing and implementing scalable, high-performance backend architectures where efficiency, correctness, and reliability are critical.

My core technical strengths lie in Python and Go, which I use for building backend services, data pipelines, and system-level components. I enjoy exploring areas such as data preparation, inference pipelines, and real-time AI processing, with a particular interest in enabling computer vision workflows, while ensuring systems operate reliably under real-world constraints such as latency, fault tolerance, and scalability.

I continuously deepen my foundation in backend engineering, algorithms, and AI systems, aiming to contribute to technically challenging projects where strong engineering principles and intelligent computer vision systems come together.

EDUCATION

Computer Programming 2023–2025
Istanbul Kultur University
GPA: 3.11 / 4.00
Relevant Courses: Data Structures, Algorithms, Database Management, Computer Vision, Programming

SKILLS

Backend & AI

- Python, Go, C#, C++, JavaScript, Node.js (Intermediate Level)
- Fundamentals of Artificial Intelligence and Machine Learning, Yolo, OpenCV, RoboFlow
- API Development and Integration

Database & System Management

- SQL (MySQL, PostgreSQL) & NoSQL (MongoDB)
- Docker and Containerization
- Linux System Administration

Version Control & DevOps
Git & GitHub, Docker

EXPERIENCE

Software Developer
Orbit Technology – Istanbul/Fatih
Nov – Present

- Contributed to the development and maintenance of scalable backend systems using Python, emphasizing modularity and clean architecture.
- Applied object-oriented programming principles to improve code maintainability and system reliability.
- Collaborated with cross-functional teams through regular meetings and discussions to optimize workflows and code quality.
- Researched ways to prepare backend systems for integration with machine learning and computer vision pipelines.

Software Developer Intern
EYTSOFT A.Ş. – Istanbul, IZU Zaim Technopark
June 2025 – August 2025

- Worked on Python OOP concepts by developing and improving algorithms.
- Developed projects using Django.
- Contributed to live projects and provided development support.
- Collaborated with teams through regular meetings and discussions.

CERTIFICATES

Go (Basic) Certificate

HackerRank

Python Programlama 401

Turkcell Academy – Gelecegi Yazarlar

JavaScript

Turkcell Academy – Gelecegi Yazarlar

MongoDB

Turkcell Academy – Gelecegi Yazarlar

HTML5

BTK Academy

Asp.Net Core MVC

Turkcell Academy – Gelecegi Yazarlar

React

Turkcell Academy – Gelecegi Yazarlar

Git and GitHub

BTK Academy

Linux and Pardus Command Line Training

Pardus – Tübitak

SQL (Intermediate) Certificate

HackerRank

PHP

BTK Academy

Node.js

Turkcell Academy – Gelecegi Yazarlar

JAVASCRIPT

BTK Academy

CSS

BTK Academy

C# ile Algoritma ve Programlama

Turkcell Academy – Gelecegi Yazarlar

Linux

Turkcell Academy – Gelecegi Yazarlar

jQuery

BTK Academy

Introduction to Artificial Intelligence

BTK Academy

LANGUAGE SKILLS

Turkish, Native

English, Intermediate (B1 – Actively improving)

PROJECTS

Global Air Pollution Monitoring Platform

Technologies: **Python (Flask/FastAPI)**, **InfluxDB**, **RabbitMQ**, **React.js**, **Mapbox**, **Recharts**, **Docker**, **Docker Compose**

A web-based platform that processes and visualizes air pollution data gathered from around the world. The system analyzes the data through scripts and alerts users when pollution levels exceed critical thresholds. The goal is to raise awareness and support data-driven decision-making for environmental issues.

<https://github.com/osmantalayhan/air-quality-app>

Smart Language Learning Assistant (In Development)

Technologies: **Python (Backend, APIs)**, **JavaScript**, **HTML**, **CSS**, **Firebase**

An interactive platform that personalizes language learning. The system provides word meanings, levels (A1-C2), pronunciation, and example sentences. Features include custom learning plans, word filtering, progress tracking, and sentence analysis for grammar and logical errors. The app offers feedback and allows users to enhance their learning through collaborative vocabulary groups.

Chat Bot – Student Affairs Q&A System

Technologies: **Python**, **Flask**, **JSON**, **Natural Language Processing (NLP)**, **FuzzyWuzzy**

Developed for the Istanbul Kültür University Student Affairs Department, this chatbot system automatically answers students' frequently asked questions. Built with Python and Flask, the bot uses a JSON dataset containing question-answer pairs and integrates fuzzy string matching (via the FuzzyWuzzy library) to detect and respond to similar queries. This approach allows the system to handle typos and near-matches effectively, enhancing user experience. The project demonstrates practical applications of NLP and AI in automating administrative support and improving digital interaction channels.

<https://github.com/osmantalayhan/chat-bot>

Advanced Blog Script

Technologies: **PHP, MySQL, JavaScript, HTML, CSS**

Developed a feature-rich blog platform using PHP and MySQL, offering functionalities such as user authentication, content management, and interactive features like comment sections and post ratings. The system includes a dynamic admin panel, responsive design, SEO optimization, and multilingual support. This project demonstrates proficiency in full-stack web development and the ability to create scalable content management systems.

<https://github.com/osmantalayhan/gelismis-blog-script>

GoLang Product Management Application

Technologies: **Go (Golang), PostgreSQL, Docker, HTML, CSS**

A product management platform developed with GoLang and PostgreSQL, leveraging Go's robust libraries and Docker for containerization. The application allows users to perform CRUD operations, manage product inventories, and interact with a PostgreSQL database via RESTful APIs. This Project demonstrates my backend development skills with Go and my experience using Docker for deployment and service management.

<https://github.com/osmantalayhan/golang-product-app>

SMS Attendance and Notification System

Technologies: **PHP, MySQL, JavaScript, HTML, CSS**

A web-based system designed to simplify attendance tracking and group notifications through SMS integration. Built using PHP and MySQL, the platform allows users to manage attendance records and send real-time updates to groups. This project provides a practical solution for educational institutions or event coordinators aiming to streamline communication and participation.

<https://github.com/osmantalayhan/sms-yoklama-bildirisi>